Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. $(2;-1)\cdot(1;2)$?

2. Does the point (1; 2) satisfy 2 $x-y=0$?

3. Solve $x+y=5,\ x-y=1.$

4. Distance from (1; 2) to (4; 6) ?

5. Complete $x^2-4x.$

6. Solve x^2+y^2 = 0 in $\mathbb{R}^2.$

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. Present the two conditions characterising a projection. Give a geometric proof and an algebraic proof for a perpendicular bisector.

Investigate before formalising

From $P(4; 3)$ and $d : y = 2x$, compare the distances to three points of d, then find and prove which one is the shortest.

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.